Assignment 2

Redesign a Notable Information Graphic

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Visualization and storytelling

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Introduction

This assignment asked us to redesign a well-known historical information graphic using modern interactive visualization tools. The goal was not necessarily to improve the original graphic, but to explore how contemporary technologies and interaction techniques could reveal new perspectives or make the visualization more engaging for a modern audience. As part of the process, we were also expected to document our design thinking, including the steps we took, the data preparation process, and the reasoning behind our design choices.

For this project, I decided to redesign the famous Russian campaign visualization created by Charles Joseph Minard. I chose this graphic mainly because I have always been interested in history since I was a child. Historical stories and events have always been fascinating to me, and Napoleon's campaign in Russia is one of the most dramatic examples of how geography, weather, and logistics can affect an army. Because of that personal interest, working with this visualization felt both engaging and meaningful.

Idea and Goal of redesign

This project focuses on redesigning the famous historical visualization created by Charles Joseph Minard, which illustrates the dramatic losses of the French army during the French invasion of Russia led by Napoleon Bonaparte. Minard's original graphic is widely recognized as one of the most powerful examples of data storytelling because it successfully combines multiple variables such as geography, army size, time, and temperature into a single visual narrative.

The main idea behind this redesign was not to replace or improve the original visualization, but rather to reinterpret it using contemporary visualization approaches while preserving its historical authenticity. One of the most important design decisions in this project was to keep the original dataset unchanged. No additional data was introduced, and the visualization was reconstructed using the same information that appears in the original graphic. This choice was made to maintain the integrity of the historical narrative and respect the originality of Minard's work.

The goal of the redesign was to explore how a classic static visualization could be translated into a modern interactive experience. While the original graphic presents the entire story in a single static image, modern visualization environments allow viewers to engage with the data more actively. By redesigning the graphic in an interactive format, the intention was to make the visualization more accessible and exploratory, allowing viewers to better understand the relationships between geography, time, and the dramatic reduction of the army.

Another objective of the redesign was to demonstrate how contemporary visualization techniques can reveal new perspectives within the same historical data. Instead of passively reading the narrative from a single image, viewers can interact with the visualization and explore different aspects of the campaign more closely. This shifts the visualization slightly from being purely rhetorical to becoming a more exploratory analytical tool.

Overall, the purpose of this redesign was to bridge historical information design with modern interactive visualization, showing how a classic work like Minard's map can be reinterpreted for contemporary audiences while still preserving its original meaning and data structure.

Data

To start the redesign process, I first carefully studied the original graphic by Charles Joseph Minard and extracted the information that was embedded in the map. Since the visualization is originally a static graphic, the data is not directly available in a dataset. Therefore, I manually reconstructed the dataset by identifying and recording the key variables shown in the map.

The variables I extracted include latitude, longitude, date, temperature, army size, city locations, and the direction of the movement (whether the army was advancing toward Moscow or retreating). After collecting all of this information, I organized the data into a structured CSV file so that it could be used in modern visualization tools. This dataset then became the foundation for building my interactive visualizations in Power BI and Flourish.

A	B	C	D	E	F	G	H
step	date	city	lat	lon	army_size	temperature_c	direction
1	1 Jun 1812	Kowno	54.9	24	422000		Advance
2	10 Jun 1812	Vilna	54.7	25.3	400000		Advance
3	15 Jul 1812	Smorgoni	54.8	26.4	340000		Advance
4	25 Jul 1812	Molodechno	54.3	26.8	320000		Advance
5	7 Aug 1812	Glubokoye	55.1	27.7	300000		Advance
6	14 Aug 1812	Polotsk	55.5	28.8	280000		Advance
7	21 Aug 1812	Witebsk	55.2	30.2	240000		Advance
8	28 Aug 1812	Smolensk	54.8	32	175000		Advance
9	7 Sep 1812	Dorogobuzh	54.9	33.3	145000		Advance
10	14 Sep 1812	Ghatsk	55.2	34.3	127000		Advance
11	21 Sep 1812	Mojaisk	55.5	35.6	110000		Advance
12	28 Sep 1812	Moscow	55.8	37.6	100000		Advance
13	10 Oct 1812	Moscow	55.8	37.6	100000	Temperature : 0	Retreat
14	20 Oct 1812	Mojaisk	55.5	35.6	98000	Temperature : -9	Retreat

Picture 1: Extracted data from Minard's visual

Flourish

Since I had never worked with Flourish before, this assignment was also a good opportunity for me to start learning and experimenting with a new visualization tool. I wanted to see how a modern interactive platform could represent the same historical data that Minard originally visualized in a static form.

In Flourish, I created three different charts based on the dataset that I reconstructed from the original map. The first two charts focus on the movement of the French army during the campaign. One chart represents the advance of the army toward Moscow, and the other shows the retreat after leaving Moscow. In these visualizations, I used a point map placed on the real geographic map of the Earth in the same region where Minard's original map is located.

Each point on the map represents a step in the army's movement. I organized the movement into 20 steps in total: steps 1 to 12 represent the forward movement toward Moscow, and steps 13 to 20 represent the retreat. To make the two phases visually clear, I used different colors. The forward movement points are shown in blue, while the retreat points are shown in pink. This makes it easier to visually distinguish between the advance and the retreat while following the path on the map.

The third chart is an interactive animation. In this visualization, the French army is represented as a moving point that travels along the path toward Moscow and then returns during the retreat. The animation allows the viewer to follow the journey step by step, which helps make the story of the campaign easier to understand and more engaging compared to a static image.

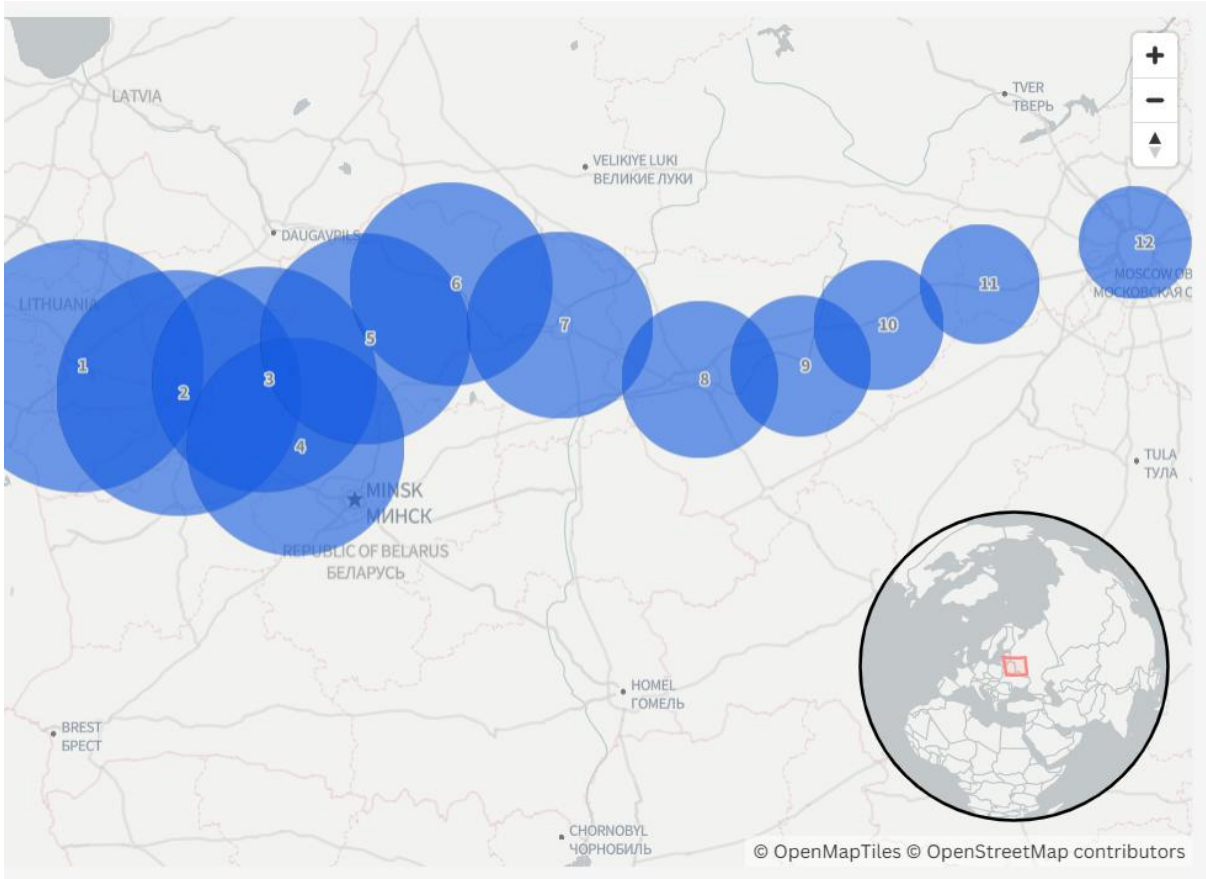

Figure 2: Flourish map depicting the French army's march toward Moscow.

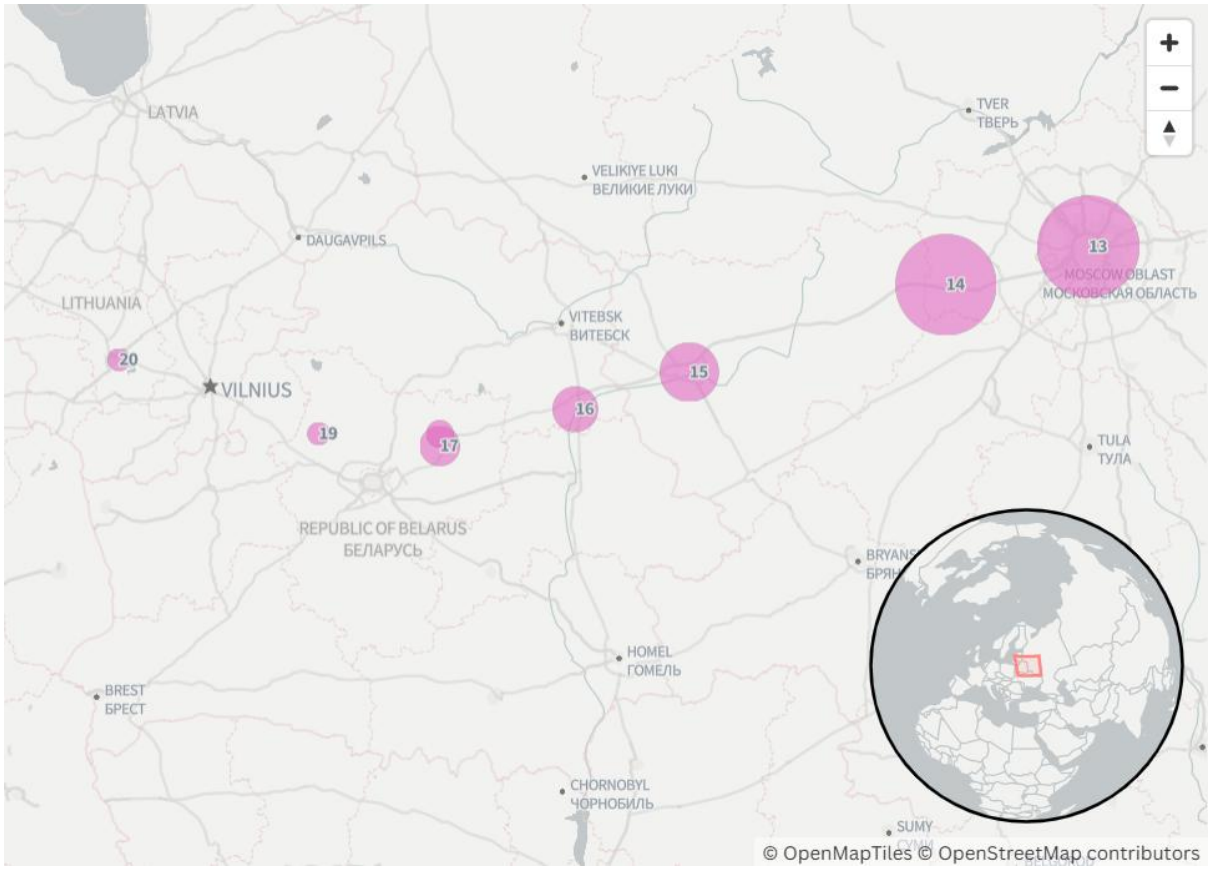

Figure 3: Flourish map visualizing the French army's march toward Moscow.

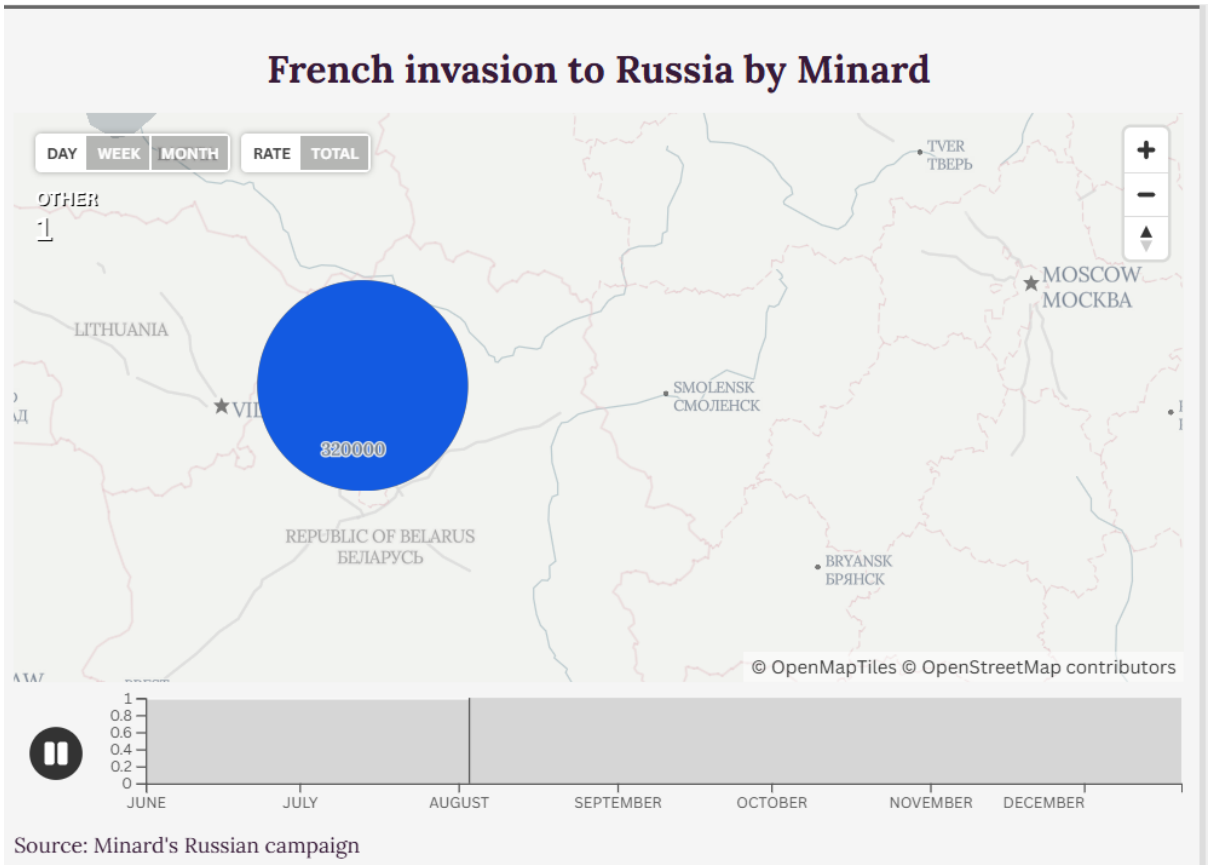

Figure 4: Interactive animation illustrating the French campaign toward Moscow.

Power BI Interactive Dashboard:

This dashboard is a data storytelling visualization built in Microsoft Power BI, inspired by the famous historical graphic created by Charles Joseph Minard that illustrates French invasion of Russia.

The goal of this project was to transform this historical dataset into an interactive analytical dashboard that allows users to explore the progression and collapse of the French army during the campaign toward Moscow and the subsequent retreat.

Map Visualization

At the center of the dashboard, I created a world map visualization that displays the geographical path of the French army during the campaign.

Each location of the army is represented as a bubble on the map, and the size of the bubble reflects the number of soldiers remaining at that point in time. This allows users to immediately see how the army size changes as it moves across different locations.

This spatial representation helps connect geography with military outcomes, showing how the army advanced deep into Russia and then rapidly declined during the retreat.

Army Size Trend Analysis

Below the map, I built a line chart that tracks the army size over time.

This visualization shows:

- The decline of the army by date
- Labels directly on the line displaying the number of soldiers at each stage
- Additional contextual information along the X-axis, including:
    1. Temperature values recorded during the retreat
    2. City locations where the army passed

This combination allows users to clearly observe the relationship between extreme temperatures and the dramatic reduction in army size, which was a key factor in the campaign's failure.

Interactive Campaign Filter

To make the dashboard more interactive, I added a slicer that allows users to filter the visualization by campaign stage:

- Advance toward Moscow
- Retreat from Moscow
- Both stages combined

This enables users to focus on a specific phase of the campaign or analyze the entire timeline together.

Key Metrics

Finally, I included two cards that summarize the scale of the campaign:

- Maximum army size
- Minimum army size

These metrics highlight the dramatic contrast between the army at the beginning of the campaign and what remained at the end.

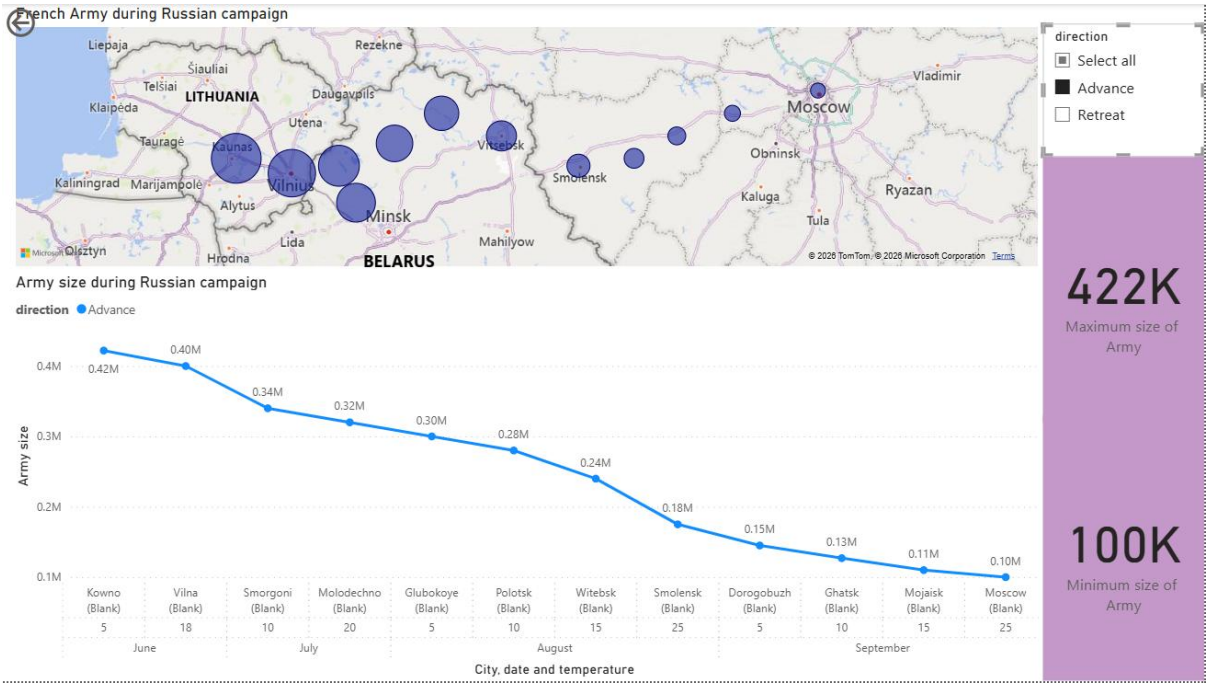

Figure 5: Interactive dashboard visualizing the advance of the French army toward Moscow.

Figure 6: Interactive dashboard visualizing the retreat of the French army from Moscow.

Conclusion

This project successfully reimagines Charles Joseph Minard's classic visualization of the French invasion of Russia using modern interactive methods. The redesign preserved the original data and historical narrative, while transforming a static graphic into an exploratory,

interactive experience that allows viewers to engage with the army's movements, size changes, and environmental conditions in new ways.

By combining interactive maps, animations, and coordinated visualizations, the project demonstrates how historical information graphics can be adapted for contemporary audiences without losing their authenticity or analytical power. Overall, this assignment highlights the potential of modern visualization tools to make complex historical data more accessible, engaging, and insightful, bridging the gap between classical design and interactive storytelling.